

Software Requirement Specifications
AI-Powered Smart University Chatbot (MYIUB)



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Meeting Details

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Summary

This document presents the Software Requirement Specifications (SRS) for the AI-Powered Smart University Chatbot that can be integrated into the MYIUB portal. The purpose of this system is to provide students and teachers with instant access to academic and administrative information using natural language queries.

Currently, users face difficulty in navigating the university portal and finding relevant information quickly. This chatbot aims to solve these problems by acting as a virtual assistant that understands user queries and provides accurate responses.

The system will use Natural Language Processing (NLP) techniques and integrate with university databases and documents to deliver both general and personalized information. The expected outcome is an efficient, user-friendly system that reduces manual effort and improves overall user experience.

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1. Introduction

The Islamia University of Bahawalpur (IUB) provides a digital portal (MYIUB) for students and teachers to access academic and administrative services. These include exam schedules, fee details,

course information, and announcements. However, users often struggle to locate relevant information due to complex navigation and lack of interactive support.

This project focuses on developing an AI-powered chatbot module that will be integrated into the MYIUB portal. The chatbot will allow users to ask questions in natural language and receive instant responses, improving accessibility and usability of the system.

1.1. Purpose

The purpose of this project is to design and develop an intelligent chatbot that:

- Provides instant responses to user queries
- Improves accessibility of university information
- Reduces manual searching effort
- Enhances user experience in the MYIUB portal

1.2. Scope

These are the scope of this project:

- Chatbot integration within MYIUB portal
- Answering academic and administrative queries
- Personalized responses (GPA, timetable, courses)
- Document-based question answering
- Role-based access (Student, Teacher, Admin)

1.3. Product Perspective

The chatbot will act as an additional module within the MYIUB portal. It will interact with existing databases and systems to retrieve information and display it in a conversational format. The system will not replace the portal but enhance its functionality.

1.4. User Characteristics

- **Students:** Basic technical knowledge, frequent users
- **Teachers:** Moderate technical knowledge, require quick access to schedules and courses
- **Admin Staff:** Advanced access for managing data and chatbot responses

1.5. Similar apps and systems/Literature Review

2. University chatbots used in international institutions
3. AI assistants like Google Assistant and ChatGPT
4. Existing portal-based search systems

Shortcomings of existing systems:

- Lack of personalization
- Keyword-based search instead of semantic understanding
- No real-time conversational support

1.6. Proposed Technologies

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python (Django/Flask)
- **AI/NLP:** NLTK or API-based models
- **Database:** MySQL / PostgreSQL
- **Integration:** REST APIs

2. Requirements

The system will provide functionalities such as answering queries, retrieving user-specific data, and accessing university documents. It will support multiple user roles and provide accurate responses based on user intent.

2.1. Function Requirements

2.1.1. Login System

- **Name:** FR001
- **Purpose:** Allow users to securely access the system
- **Users:** Admin, Student, Teacher
- **Input:**
 - Username
 - Password
- **Output:** User will be successfully able to login

2.1.2. Chat Query Processing

- **Name:** FR002
- **Purpose:** Process user queries using NLP
- **Users:** Student, Teacher
- **Input:**
 - Natural language query

- **Output:** Relevant response displayed

2.1.3. Personalized Information

- **Name:** FR003
- **Purpose:** Provide user-specific data
- **User:** Student
- **Input:**
 - Logged-in user request
- **Output:** GPA, timetable, courses

2.1.4. Document-Based Q&A

- **Name:** FR004
- **Purpose:** Retrieve answers from university documents
- **Users:** Admin, Student, Teacher
- **Input:**
 - Query related to policies
- **Output:** Extracted relevant information

2.1.5. Admin Management

- **Name:** FR005
- **Purpose:** Manage chatbot data and responses
- **Users:** Admin
- **Input:**
 - Username
 - New data entries (FAQs, policies, responses)
 - Updated academic information (schedules, deadlines)
 - User feedback and corrections
 - Document uploads (PDFs, notices, policies)
 - Commands to modify or delete existing data
- **Output:**
 - Updated system knowledge base
 - Improved and accurate responses for users

2.2. Non-Functional Requirements

- **Performance:** Fast response time (<2 seconds)
- **Security:** Secure login and data protection
- **Usability:** Easy-to-use chat interface
- **Scalability:** Can handle multiple users
- **Reliability:** System available 24/7
- **Accuracy:** High response accuracy using NLP

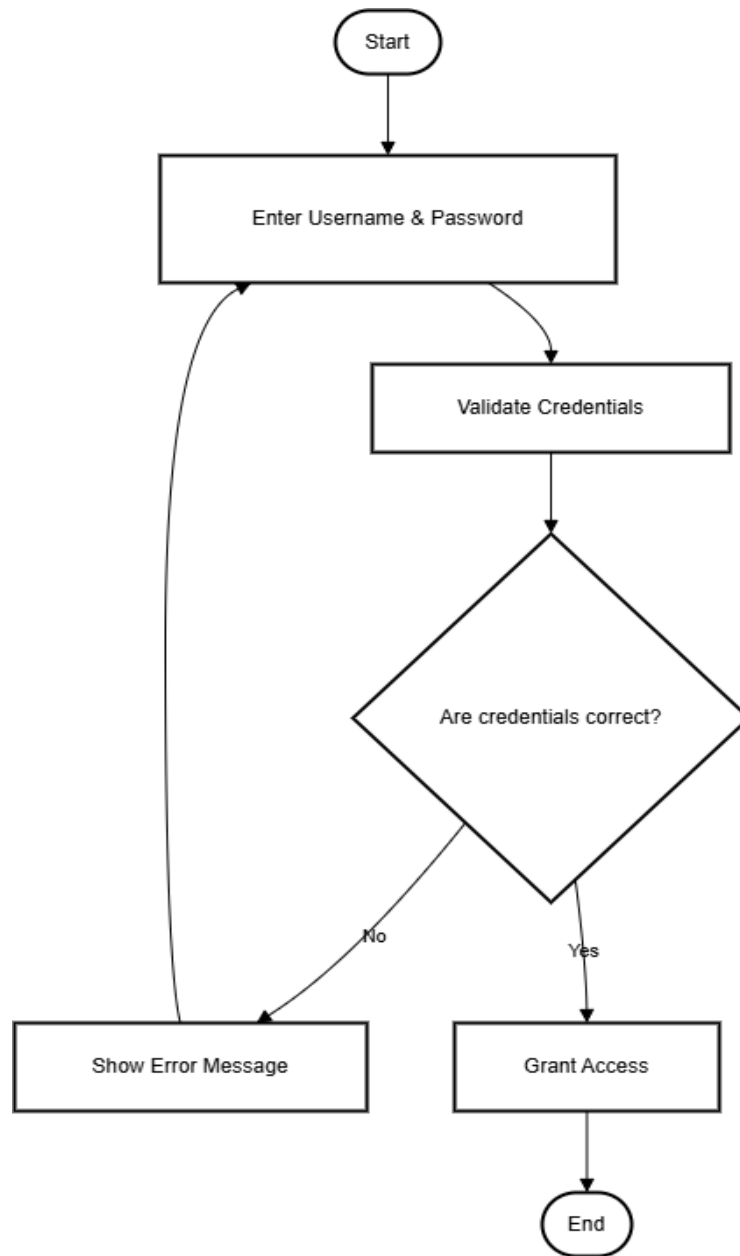
3. Use Cases and Flow of Processes

The system will include multiple use cases such as login, query processing, and data retrieval.

3.1. Use Case 1

ID	UC001
Name	User Login
Description	User logs into the MYIUB chatbot system
Requirement(s)	FR001
Actor(s)	Student, Teacher, Admin
Precondition	User must be registered
Postcondition	User is logged into system
Basic Flow	<i>Basic Flow</i> 1. User enters username and password 2. System validates credentials 3. If correct, access is granted <i>Alternative Flow</i> • Incorrect credentials → error message <i>Exceptions</i> • System error • Server not responding

Diagram:

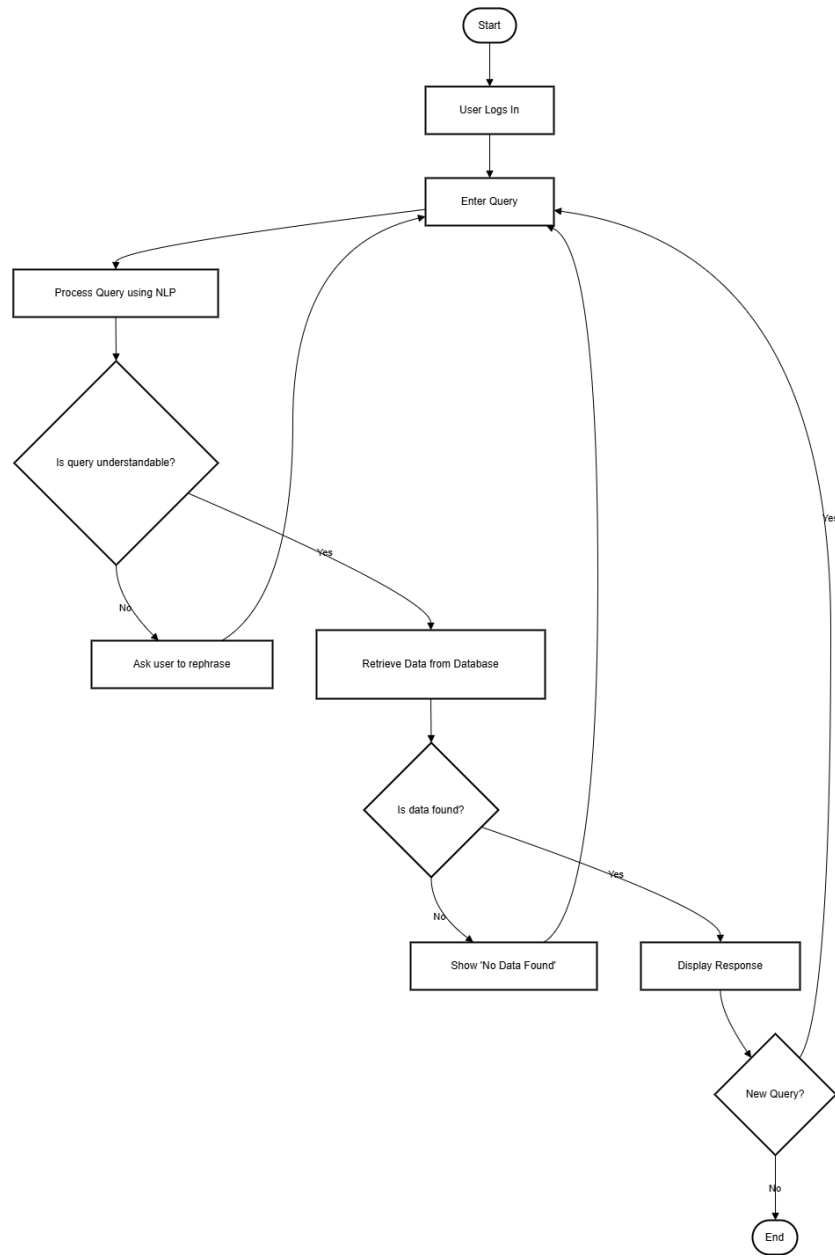


3.2. Use Case 2

ID	UC002
Name	Ask Query
Description	User asks a question to chatbot
Requirement(s)	FR002

Actor(s)	Student, Teacher
Precondition	• User must be successfully logged into the MYIUB portal • Chatbot system must be active and connected to database • User must have appropriate role (Student/Teacher/Admin)
Postcondition	• User receives a relevant response from the chatbot • Query is processed and optionally stored for future improvement • System logs the interaction for analytics and learning
Basic Flow	<i>Basic Flow</i> 1. User enters query 2. System processes using NLP 3. System retrieves data 4. Response displayed <i>Alternative Flow</i> • Invalid or Unclear Query • No Data Found • Personalized Data Access Failure • System Processing Delay <i>Exceptions</i> • Invalid query • No data found • AI service unavailable

Diagram:

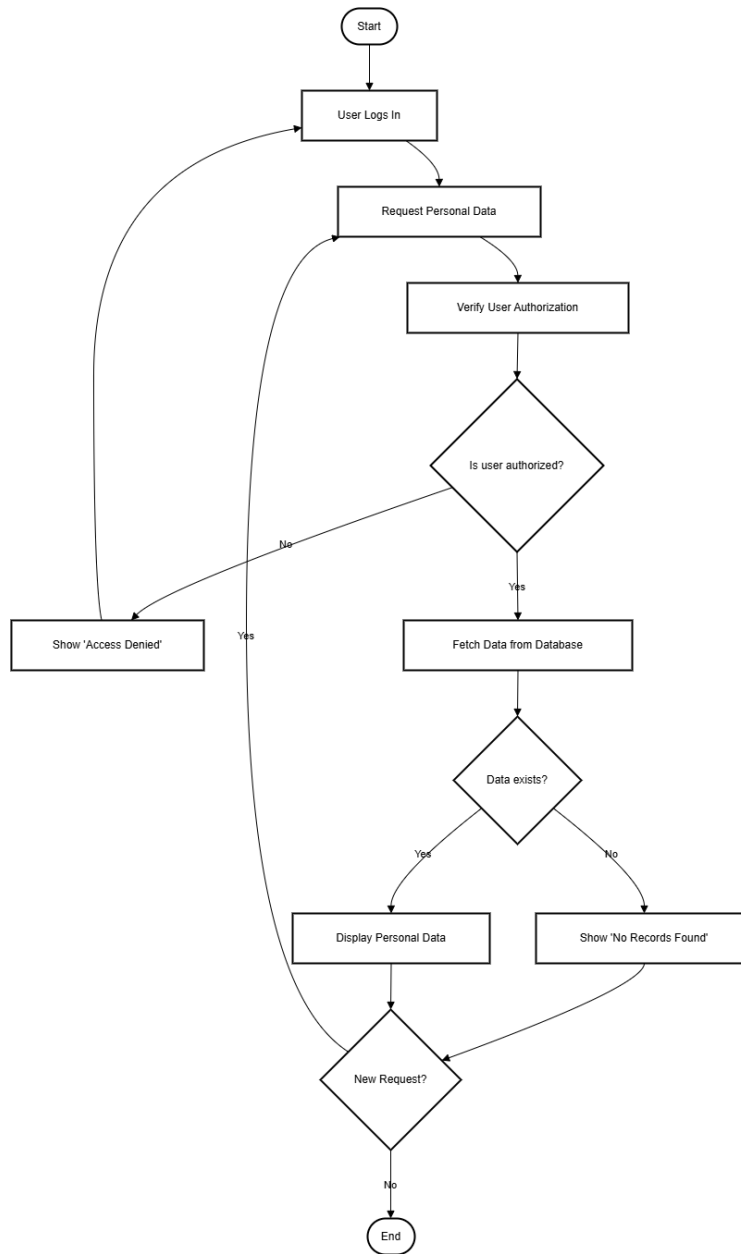


3.3. Use Case 3

ID	UC003
Name	View Personal Data
Description	This use case allows users to view their stored profile information
Requirement(s)	FR003
Actor(s)	Student, Teacher, Admin

Precondition	User must be logged into the system
Postcondition	User can view personal profile details
Basic Flow	<i>Basic Flow</i> 1. User selects “Profile” option 2. System retrieves user data from database 3. System displays personal information <i>Alternative Flow</i> • User requests data update → redirected to edit profile <i>Exceptions</i> • Data not found • Unauthorized access attempt

Diagrams

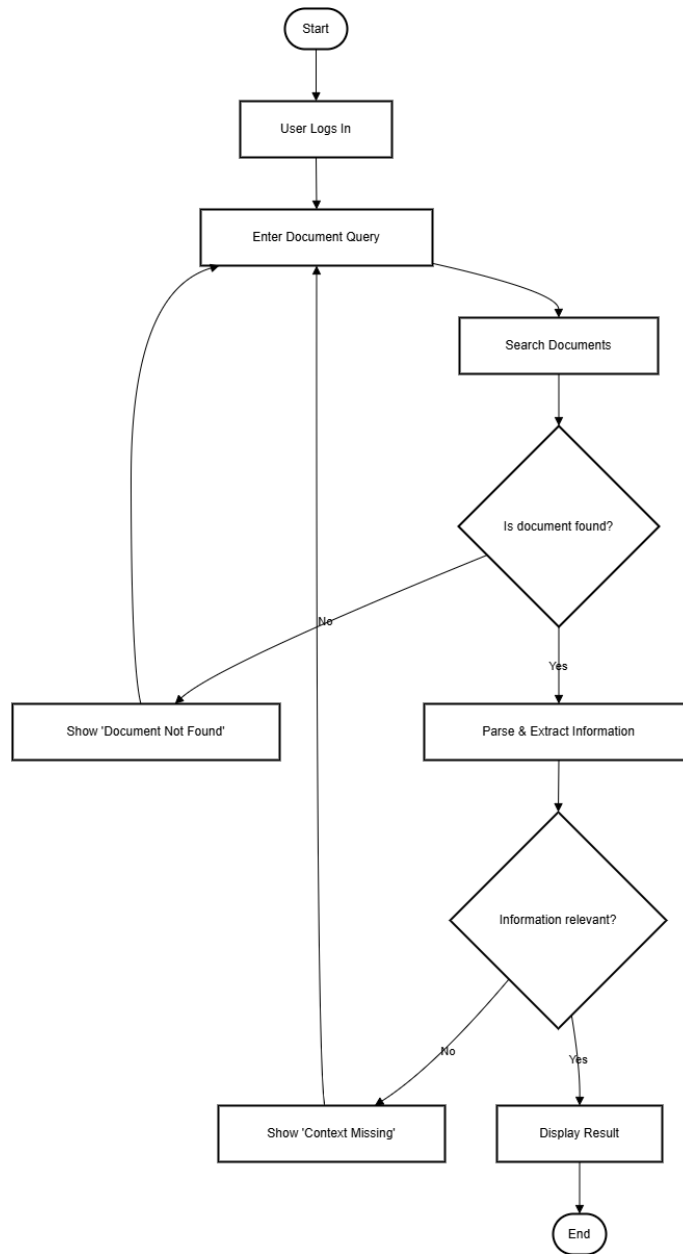


3.4. Use Case 4

ID	UC004
Name	Access Documents
Description	This use case describes how users can access uploaded or stored academic documents.
Requirement(s)	FR004

Actor(s)	Student, Teacher, Admin
Precondition	User must be authenticated
Postcondition	User can view or download documents
Basic Flow	<i>Basic Flow</i> 1. User opens document section 2. System lists available documents 3. User selects a document 4. System opens or downloads file <i>Alternative Flow</i> • Filter documents by category (notes, assignments, etc.) <i>Exceptions</i> • File not found • Access denied

Diagrams

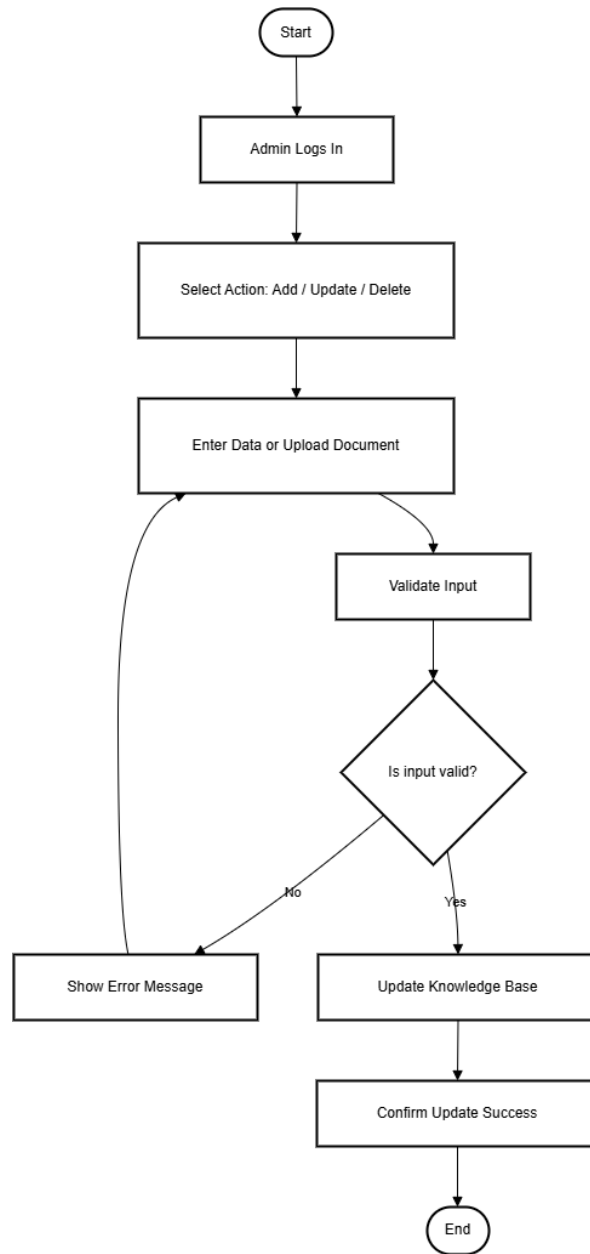


3.5. Use Case 5

ID	UC005
Name	Admin Manage Data
Description	This use case describes how admin manages system users and data.
Requirement(s)	FR005
Actor(s)	Admin

Precondition	Admin must be logged in
Postcondition	System data is updated successfully
Basic Flow	<i>Basic Flow</i> 1. Admin accesses management dashboard 2. Admin views user/data records 3. Admin adds, updates, or deletes data 4. System saves changes <i>Alternative Flow</i> • Admin approves/rejects user accounts <i>Exceptions</i> • Unauthorized access • Database error

Diagrams



4. References

- IEEE SRS Documentation Standards
- Research papers on AI Chatbots
- NLP documentation (NLTK, SpaCy)
- MYIUB portal system references